Term 3 answer key

Formulas from memory. Lengths, areas and volumes are rounded only where the question says so (1 decimal place, the nearest degree or the nearest cent).

WEEK	SESSION	Q1 A Triangles & similarity	Q2 B Area & volume	Q3 C Percentages & money	Q4 D Indices, formulas & rates	Q5 E Graphs, data & chance
1	1	8.8 cm	7.9 cm	$1750	5^6	10
	2	23°	197.9 cm^3	8% loss	130 km/h	4.1
	3	8.3 cm	72 L	$441	9×10^{-1}	0.65
2	1	6 cm	414.7 cm^2	$78	$140, $210 and $280	25
	2	4/5	70 cm^3	$1222	$1/7^3$ = 1/343	3/8
	3	11.3 cm	67 cm^2	$7050	23	1
3	1	7.5 cm	63.6 m^2	$936	$80, $120 and $160	3/5
	2	34°	113.1 cm^2	$17 125	4.5 hours	gradient −4, *y*-intercept 1
	3	4/3	7.6 L	$240	1	−1/2
4	1	8.5 m	98 cm^2	$5938.45	6	0.25 or 1/4
	2	13.2 cm	7 m	$773	0.000016	9.4
	3	10.0 cm	11.8 cm	$2855.77	$14a^7$	5/12
5	1	7.5 cm	235.6 m^3	$500	$60 and $90	0.53
	2	32.7 m	285.9 cm^2	$71.96	26	0.175 or 7/40
	3	28°	227.0 cm^2	$896.80	1/1000	1/2
6	1	25 cm	42.4 cm^2	25% profit	8 m/s	7.1
	2	4.1 cm	502.7 m^3	$1591.35	6.2×10^{-3}	1/2
	3	7.5 cm	13.9 L	$5640	$3m^5$	gradient −2, *y*-intercept 6
7	1	4/5	628.3 cm^2	$80	6.4	1/3
	2	9 cm	14.1 m^2	$1475.80	3/4 or 0.75	50
	3	26.9 cm	15.7 m	$1650	$27a^6$	3
8	1	54°	127.5 cm^3	20% loss	8 m/s	0.18
	2	15.3 cm	50.3 cm^2	$340	$210 and $150	3/2
	3	5/13	94 cm^2	$825	0.0308	69
9	1	12.6 cm	64.1 m^2	$360	1	7/12
	2	63.8 m	445.3 cm^3	$80	90 km/h	3/7
	3	2 cm	245.0 cm^2	25% profit	$3y^5$	gradient −2, *y*-intercept 7
10	1	12.2 cm	81 cm^3	$4629.50	1.5×10^5	3
	2	14.2 cm	22.0 cm	$750	5	0.36 or 9/25
	3	9.2 m	63.6 m^2	$3665.38	$40 and $160	10

NSW MATHEMATICS K–10 (2022) OUTCOMES PRACTISED · STAGE 5, WITH STAGE 4 REVISION · MAO-WM-01 THROUGHOUT

A MA4-PYT-C-01 Pythagoras' theorem; MA5-TRG-C-01 trigonometric ratios (sides and angles); MA5-TRG-C-02 angles of elevation and depression (Terms 3–4); MA5-GEO-C-01 similar figures

B MA4-ARE-C-01 area of circles and sectors*; MA4-LEN-C-01 arc length*; MA5-ARE-C-01 surface area of prisms and cylinders; MA5-VOL-C-01 volume of prisms and cylinders, capacity; MA5-VOL-P-01 (Path, Term 4 only) volume of pyramids and cones

C MA5-FIN-C-01 simple interest, earning money (wages, overtime, salary, commission, leave loading); MA4-FRC-C-01 percentage increase and decrease, original amount, percentage profit and loss*

D MA5-IND-C-01 index laws with coefficients, zero and negative indices (numerical bases); MA5-MAG-C-01 scientific notation; MA4-ALG-C-01 / MA5-EQU-C-01 substituting into formulas and solving*; MA4-RAT-C-01 ratios and rates (speed)

E MA5-LIN-C-01 gradient and $y = mx + c$; MA4-DAT-C-02 mean, including frequency tables*; MA4-PRO-C-01 / MA5-PRO-C-01 probability, complementary events, relative and expected frequency*

* the code is right for the topic, but where this content sits within it could not be checked against the NESA website.